

An Asymptotic-Dimension-Zero Counterexample for Asymptotic Coarse Lipschitz Equivalence

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Source Question

The source is Bruno M. Braga and Gilles Lancien, *Asymptotic coarse Lipschitz equivalence*, arXiv:2302.12016. The paper introduces asymptotic coarse Lipschitz equivalence: metric spaces X and Y are asymptotically coarse Lipschitz equivalent if there are coarse Lipschitz maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ such that gf and fg are asymptotically close to the corresponding identity maps.

After constructing examples separating asymptotic coarse Lipschitz equivalence from coarse equivalence in positive asymptotic dimension, the paper asks:

Problem 2.15. Let X and Y be asymptotically coarse Lipschitz equivalent metric spaces with asymptotic dimension zero. Does it follow that X and Y are coarse (Lipschitz) equivalent?

It is easy to show that this is a coarse property, i.e., it is preserved by coarse equivalences. But while X in the previous example has it, Y does not.

Notice that both restrictions in the previous proposition and in the previous remark only work for metric spaces with asymptotic dimension at least 1 (indeed, the existence of the paths in the remark above implies either that X is bounded or that X has asymptotic dimension at least 1). Moreover, Example 2.9 falls in the same scenario. This justifies the following problem.

Problem 2.15. Let X and Y be asymptotically coarse Lipschitz equivalent metric spaces with asymptotic dimension zero. Does it follow that X and Y are coarse (Lipschitz) equivalent?

Answer

The answer is negative as stated. We construct two metric spaces X and Y with asymptotic dimension zero which are asymptotically coarse Lipschitz equivalent but are not even coarsely equivalent.

Definitions Used

For maps $u, v : Z \rightarrow W$, write $u \sim_\infty v$ if, for some equivalently every base point $z_0 \in Z$,

$$\lim_{z \rightarrow \infty} \frac{d_W(u(z), v(z))}{d_Z(z, z_0)} = 0.$$

A map $h : Z \rightarrow W$ is coarse Lipschitz if its modulus of continuity is bounded above by an affine function. A pair of coarse Lipschitz maps $f : X \rightarrow Y$, $g : Y \rightarrow X$ witnesses asymptotic coarse Lipschitz equivalence if $gf \sim_\infty \text{Id}_X$ and $fg \sim_\infty \text{Id}_Y$.

We use the standard characterization of asymptotic dimension zero: a metric space has asymptotic dimension zero if, for every $r > 0$, the r -chain components have uniformly bounded diameter. In the proof below this condition is verified directly.

Proof Intuition

Start with a very sparse countable metric space $X = \{x_n\}$, where different points are separated by distances on the order of $2^n + 2^m$. Build Y by replacing x_n with an uncountable discrete fiber of diameter n . The fiber diameter n is negligible compared with the distance 2^n from the base point, so collapsing each fiber gives an asymptotic coarse Lipschitz inverse. But a genuine coarse equivalence would have to coarsely cover each far fiber within one fixed radius. For a fiber whose internal distance is larger than that radius, every small ball is a singleton, so a countable space cannot coarsely cover an uncountable fiber.

The Counterexample

Theorem 1. *There exist metric spaces X and Y such that*

$$\text{asdim } X = \text{asdim } Y = 0,$$

X and Y are asymptotically coarse Lipschitz equivalent, but X and Y are not coarsely equivalent. In particular, they are not coarse Lipschitz equivalent.

Proof. Let A be an uncountable set and fix $a_0 \in A$. Let

$$X = \{x_n : n \in \mathbb{N}\}$$

with metric

$$d_X(x_n, x_m) = \begin{cases} 0, & n = m, \\ 2^n + 2^m, & n \neq m. \end{cases}$$

Let

$$Y = \mathbb{N} \times A$$

with metric

$$d_Y((n, a), (m, b)) = \begin{cases} 0, & (n, a) = (m, b), \\ n, & n = m \text{ and } a \neq b, \\ 2^n + 2^m, & n \neq m. \end{cases}$$

The triangle inequality is immediate: cross-fiber distances are star distances $2^n + 2^m$, while an internal fiber distance n is at most $2^n + 2^n$, so passing through another fiber cannot shorten it.

We first check asymptotic dimension zero. Fix $r > 0$, and choose N so large that

$$N > r \quad \text{and} \quad 2^N > r.$$

In X , all points x_n with $n \geq N$ are isolated in the r -chain relation, while the remaining points form only finitely many r -components, all with bounded diameter. Thus X has asymptotic dimension zero.

The same argument applies to Y . If $n \geq N$, then distinct points in the fiber $\{n\} \times A$ are at distance $n > r$, and points in different sufficiently large fibers are also more than r apart. All possible nontrivial r -components lie over the finite set $\{1, \dots, N-1\}$, hence have diameter bounded by a finite constant depending only on r . Therefore Y has asymptotic dimension zero.

Define

$$f : X \rightarrow Y, \quad f(x_n) = (n, a_0),$$

and

$$g : Y \rightarrow X, \quad g(n, a) = x_n.$$

The map f is an isometric embedding, and g is 1-Lipschitz: distances between different fibers are preserved, while distances inside a fiber are collapsed to zero. Hence both maps are coarse Lipschitz. Moreover,

$$gf = \text{Id}_X.$$

For $y = (n, a) \in Y$,

$$d_Y(fg(y), y) = d_Y((n, a_0), (n, a)) \leq n.$$

Taking $y_0 = (1, a_0)$, we have $d_Y((n, a), y_0) = 2^n + 2$ whenever $n \neq 1$. Thus

$$\frac{d_Y(fg(n, a), (n, a))}{d_Y((n, a), y_0)} \leq \frac{n}{2^n + 2} \rightarrow 0$$

as $y \rightarrow \infty$. Hence $fg \sim_\infty \text{Id}_Y$, and f, g witness an asymptotic coarse Lipschitz equivalence between X and Y .

It remains to prove that X and Y are not coarsely equivalent. Suppose, toward a contradiction, that there are coarse maps $F : X \rightarrow Y$ and $G : Y \rightarrow X$ such that

$$FG \sim \text{Id}_Y.$$

Then there is $C < \infty$ such that

$$d_Y(FG(y), y) \leq C \quad (y \in Y).$$

Choose $n > C$. For every $a \in A$, the closed C -ball around (n, a) is the singleton $\{(n, a)\}$: any other point in the same fiber is at distance $n > C$, and every point in a different fiber is even farther away. Therefore

$$FG(n, a) = (n, a) \quad (a \in A).$$

This implies that the image $F(X)$ contains the whole fiber $\{n\} \times A$. But X is countable, so $F(X)$ is countable, whereas $\{n\} \times A$ is uncountable. This is impossible. Hence no coarse equivalence exists. $\square$

Novelty and Duplicate Checks

Before promotion, I searched the cheap run indexes for arXiv 2302.12016, the title *Asymptotic coarse Lipschitz equivalence*, and core phrases including *asymptotic coarse Lipschitz*, *asymptotic dimension zero*, *coarse Lipschitz equivalent*, and Problem 2.15. No existing packet or attempt answered this problem.

I also ran bounded web searches for exact and near-exact phrases including

- "Asymptotic coarse Lipschitz equivalence" "Problem 2.15",
- "asymptotic coarse Lipschitz equivalent" "asymptotic dimension zero",
- "asymptotically coarse Lipschitz equivalent" "coarse Lipschitz equivalent".

No later answer surfaced in this search; the only direct hit was the source paper itself.

Scope and Limitations

The counterexample answers Problem 2.15 as written for arbitrary metric spaces. It is nonseparable and uses uncountable fibers. It does not settle strengthened variants in which the metric spaces are required to be countable, separable, uniformly locally finite, of bounded geometry, or otherwise cardinal-controlled.

References

References

- [1] B. M. Braga and G. Lancien, *Asymptotic coarse Lipschitz equivalence*, arXiv:2302.12016, 2023.

Human Review Recommendation

Review should focus on whether the source problem intended hidden separability or bounded geometry assumptions. Without such additional hypotheses, the construction gives a direct negative answer: the asymptotic error is sublinear along each uncountable fiber, while coarse equivalence would force a countable image to cover an uncountable far fiber at bounded scale.

Human Review: Asymptotic Dimension Zero ACL Counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 19 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proposed construction gives a correct negative answer to Problem 2.15 of Braga–Lancien, *Asymptotic coarse Lipschitz equivalence*, as the problem is stated for arbitrary metric spaces. The source question asks whether two asymptotically coarse Lipschitz equivalent metric spaces of asymptotic dimension zero must be coarse, or coarse Lipschitz, equivalent. The packet constructs spaces X and Y with

$$\text{asdim}(X) = \text{asdim}(Y) = 0,$$

which are asymptotically coarse Lipschitz equivalent but are not even coarsely equivalent. Since coarse Lipschitz equivalence implies coarse equivalence, this is the stronger negative conclusion.

The idea is clear. The space $X = \{x_n : n \in \mathbb{N}\}$ is a sparse sequence with

$$d_X(x_n, x_m) = 2^n + 2^m \quad (n \neq m).$$

The space Y replaces each point x_n by a fiber $\{n\} \times A$, with internal distance n , while keeping different fibers at distance $2^n + 2^m$. Collapsing each fiber gives a coarse Lipschitz map $g : Y \rightarrow X$, and choosing one representative in each fiber gives a coarse Lipschitz map $f : X \rightarrow Y$. One has $gf = \text{Id}_X$, while fg moves a point in the n -th fiber by at most n . Since this is sublinear compared with the distance scale 2^n , the maps witness asymptotic coarse Lipschitz equivalence.

The asymptotic dimension zero verification is also sound. At a fixed scale r , the finitely many small fibers may be grouped into one bounded set, while for all sufficiently large n , the points inside the n -th fiber are already farther than r apart, so singletons handle the large fibers.

The proof that the spaces are not coarsely equivalent is correct. If a coarse equivalence existed, there would be coarse maps $F : X \rightarrow Y$ and $G : Y \rightarrow X$ with FG uniformly close to Id_Y . Thus there is $C < \infty$ such that

$$d_Y(FG(y), y) \leq C \quad (y \in Y).$$

Choosing a fiber whose internal distance is larger than C forces FG to be the identity on that fiber. In the packet's original version, where the fiber is uncountable and X is countable, this gives an immediate cardinality contradiction.

Suggested Countable Variant

The uncountability of the set A is not actually needed. A more natural version of the example is obtained by taking

$$A = \mathbb{N},$$

so that both X and $Y = \mathbb{N} \times \mathbb{N}$ are countable. The asymptotic coarse Lipschitz equivalence and the asymptotic-dimension-zero argument are unchanged.

Only the final non-coarse-equivalence argument needs a small adjustment. If $F : X \rightarrow Y$ and $G : Y \rightarrow X$ were coarse maps with $FG \sim \text{Id}_Y$, let C be the closeness constant and choose $n > C$. Then FG must be the identity on the infinite fiber

$$Y_n = \{n\} \times \mathbb{N}.$$

Hence

$$F(G(Y_n)) = Y_n.$$

But Y_n has finite diameter n . Since G is coarse, $G(Y_n)$ has bounded diameter in X . Every bounded subset of the sparse space X is finite, so $G(Y_n)$ is finite, and therefore $F(G(Y_n))$ is finite. This contradicts the infinitude of Y_n .

Thus the example can be made countable. It still does not have bounded geometry or locally finite fibers, so it should still be viewed as answering the literal formulation of the source problem rather than any stronger bounded- geometry version.

Review Outcome

Archive this packet as a correct literal-wording counterexample to Problem 2.15. For presentation, I recommend replacing the uncountable fiber set by the countable choice $A = \mathbb{N}$, together with the adjusted finite-image argument above. This makes the example more natural while preserving the same obstruction.